

2(a)(i)	any three from: 1 environmental / not genetic ; 2 as change in yield occurs in one goat ; 3 yield increases (from 1st to 2nd) in all genotypes ; 4 older / 2nd lactation, goats have, bigger / more developed, udders ; 5 age / maturity, is an environmental factor ; 6 AVP ;	3
2(a)(ii)	any three from: 1 (substitution / G → C / CC) increases milk, yield / production ; 2 more food / more income / benefit for people ; 3 little difference between GG and CG / small effect in heterozygotes ; 4 AVP ;	3
2(b)(i)	CC = $(0.3 \times 0.3 \times 268 =)$ 24 ; CG = $(2 \times 0.3 \times 0.7 \times 268 =)$ 113 ;	2
2(b)(ii)	chi-square(d test) / χ^2 ;	1
2(b)(iii)	1 CC, higher<u>er</u> than predicted / 47 instead of 16 or GG, higher<u>er</u> than predicted / 324 instead of 289 or CG, lower<u>er</u> than predicted / 69 instead of 135 ; plus any two from: 2 could be (natural / artificial) selection against heterozygotes ; 3 could be (artificial) selection for, high milk yield / CC ; 4 migration / immigration / emigration ; 5 genetic drift ; 6 non-random mating ;	3

Question	Answer	Marks
2(c)	<i>advantage:</i> 1 treat (genetic) disease / treat people / reduce suffering / therapeutic ; <i>problem (max 1):</i> 2 possible side effects (on human) / drug may not be pure / may contain allergens / risk of harm / may be unsafe / risk of goat disease transmission to humans ; 3 process may harm goats / animal welfare issues ; 4 vegetarians / vegans, may not take drug ;	2